Heat Transfer in TPMS-structures

Final Presentation

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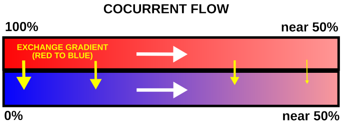
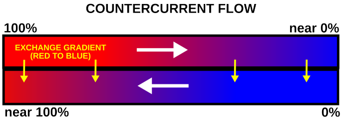
Heat Exchangers

Physical models:

- Fluid dynamics (Navier-Stokes eqns.)
- Heat conduction (Heat eqn.)





Introduction

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- Heat exchangers are traditionally evaluated by use of a body-fitted mesh
- For complex geometries, this is unfeasible
- Warrants the investigation into alternative methods (Brinkman's method)
- Much of the theory is based on the paper [YJW+23] by Jiang et al.

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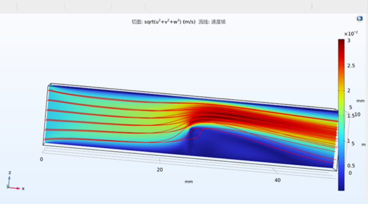
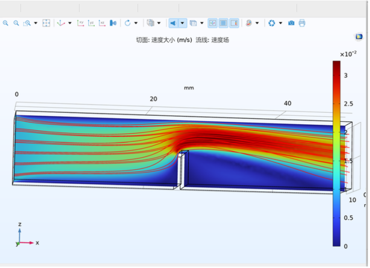
Fluid flow

Initially we implemented a simple model for fluid flow using the Brinkman method (left fig.), comparing it to the same geometry evaluated using traditional methods (right fig.).

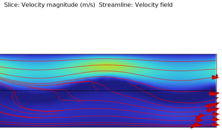


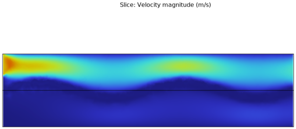


Note: All figures in this presentation are slices from 3D models.

Channel separation

Critical to the performance of the model is that the two channels are adequately separated, i.e. that there is no flow between the two fluids.
Simply implementing the Brinkman method 'as-is' does not separate the channels enough.





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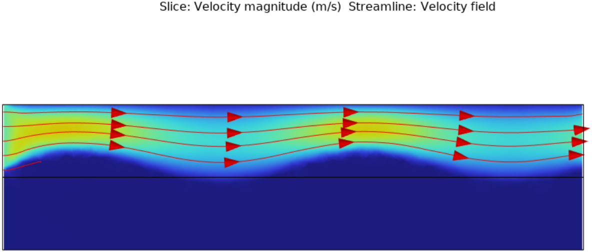
Solid region: $|z - 0.05 - 0.01\sin(10\pi x)| < 0.005$

Single channel

For a single channel we can define *both* the wall and other channel so behave like solids, by changing the solid region to:

$$z - 0.05 - 0.01\sin(10\pi x) < 0.005.$$

This renders:



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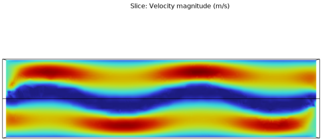
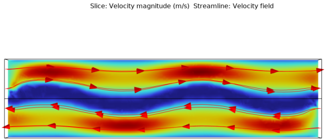
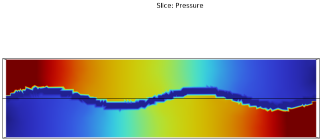
Dual channels

Luckily the Comsol framework can be nicely adjusted to incorporate two channels in the same model:





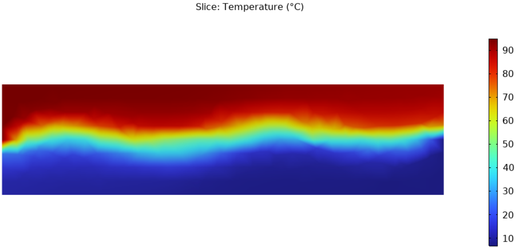


$$z - 0.05 - 0.01\sin(10\pi x) < 0.005 \qquad \text{(Upper channel)}$$

$$z - 0.05 - 0.01\sin(10\pi x) > -0.005 \qquad \text{(Lower channel)}$$

Heat transfer

Using the same model as before we can turn on the *Heat Transfer in Fluids* module and solve for the temperature distribution. The thermal conductivity is set to $0.6\,\mathrm{W\,m^{-1}\,K^{-1}}$ for water and $400\,\mathrm{W\,m^{-1}\,K^{-1}}$ for copper. From this we can see that the current design of the heat exchanger is very poor:



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Background

Triply Periodic Minimal Surfaces (TPMS) incorporates a wide range of surfaces, of which the *gyroid* is the subject of this project. Discovered by NASA scientist Alan Schoen in 1970, it was proven to be embedded by Grosse-Brauckmann and Wohlgemuth in 1996 [Wik26].

- Embedded; divides space into two *disjoint* sub-volumes.

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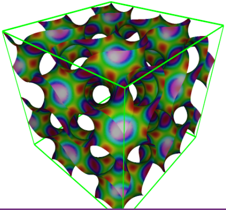

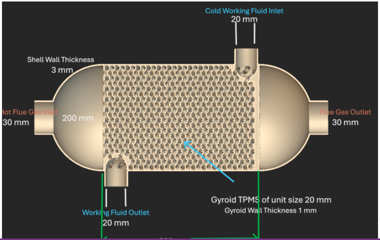

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- Geometry can be fine-tuned by function-based parameters.
- *Disadvantage:* Difficult to simulate performance

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[YJW+23] Jiang Yu, Hu Jiangbei, Shengfa Wang, Na Lei, Zhongxuan Luo, and Ligang Liu.
Meshless optimization of triply periodic minimal surface based two-fluid heat exchanger.
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